

Bellows Plus Requirements

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September 21, 2024**

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1 Introduction

The purpose of this document is to define the requirements for bellows plus. Bellows plus (BP) is an extension of bellows, a toolset that provides an interface to ZigBee hardware components. This extension provides a gateway server to bellows so that ZigBee devices that primarily have on/off switches can be accessed with simple endpoint referenced on, off, and status commands. Tools in BP will use the ZigBee database to build a template web server for direct user on/off commands to ZigBee devices and a timed event generator to control the sequencing of these devices based on time or time related events.

1.1 Scope

This document defines the interface and processing for the BP gateway, BP http-server, and BP timed event generator.

1.2 Features

The BP gateway will connect to bellows and provide a web socket interface to other entities

The BP http-server will provide a html interface to a user so that a browser can turn on and off ZigBee devices by pressing on/off buttons displayed on a browser. The http-server will send ZCL commands to bellows via the gateway and use the status returned to display device states.

The BP timed event generator will turn on and off devices based on time or time related events (eg sunrise or sunset).

BuildTools will build the basic utilities listed in this paragraph using the ZigBee database.

1.3 Acronyms

BP	
HTTP	
UTC	
ZCL	

2 General Overview

BP will consist of a set of templates and a BuildTools utility. The utility will be used along with a ZigBee database to create a gateway, HTTP server, and timed event generator that support ZigBee devices in the network specified by the database.

3 BP

The following paragraphs outline the operation of BP and define the requirements of its behavior.

3.1 Theory of Operation

BP tools will run on a Debian platform such as Raspberry Pi OS. BuildTools will be initiated from the command line of a shell and configure the gateway, http-server, and timed event generator based on the contents of the ZigBee database. If, for instance, the database contains five devices that have on/off switches, the gateway template will be used to make a gateway that will send on or off commands to the five different devices. The http-server template will be used to configure ten buttons to control the five devices. The timed event generator will be configured to turn on and off the five different devices. The user can then hand edit the http-server and the timed event generator to name the devices and set the desired times for the desired devices to be turned on or off.

Once configured, the gateway, http-server, and timed event generator can be started to do their respective jobs.

The http-server will display a single page of buttons when a browser connects to its http port. When a user presses a button, the action represented by the button (eg turn on a device or turn off a device) will cause a command to be sent to the gateway.

The gateway will use the implied action to command bellows to perform the action. The gateway will monitor the network status and send status information to any web sockets connected to it.

The timed event generator will send on/off commands to devices in the network based on time. Time will include sunrise and sunset events and offsets from those events. The server will run independently of the gateway and http-server but will use the gateway web socket to perform device commands in the same way the http-server does.

3.2 External Interface Requirements

ID::BP-EXT-IF-010 The BP gateway shall use the bellows ZigBee database to access the ZigBee devices.

ID::BP-EXT-IF-020 The BP gateway shall use bellows to interact with the devices.

ID::BP-EXT-IF-030 An HTTP server shall provide the interface to the user for turning on and off devices.

ID::BP-EXT-IF-041 The BP timed event generator shall use sunrise and sunset information provided in configuration files. This information will specify the UTC time for sunrise and sunset on each day of the year where the event or events occur. Daylight savings time will not be used in the timed event generator.

3.3 Processing Requirements

The following paragraphs identify the processing to be performed by the components of BP.

3.3.1 BuildTools

ID::BP-PR-BTGW-010 BuildTools shall configure the BP gateway so that it can communicate to all of the ZigBee devices listed in the ZigBee database that have on/off clusters and on/off status attributes.

ID::BP-PR-BTHS-010 BuildTools shall configure the http-server as specified in the following paragraphs.

ID::BP-PR-BTHS-010.001 BuildTools shall configure http-server to have on/off buttons for every device in the ZigBee database that has on/off clusters.

ID::BP-PR-BTHS-010.002 BuildTools shall configure http-server to support on/off commands that instruct the gateway as to the device and the action (on/off) to perform.

ID::BP-PR-BTTS-010 BuildTools shall configure the timed event generator as specified in the following paragraphs.

ID::BP-PR-BTTS-010.001 BuildTools shall configure the timed event generator to support on/off commands that instruct the gateway as to the device and the action (on/off) to perform.

ID::BP-PR-BTTS-010.002 BuildTools shall configure the timed event generator to support the following time events:

ID::BP-PR-BTTS-010.002.001 On time, turn on. At a specified time in UTC, turn on a device.

ID::BP-PR-BTTS-010.002.002 On time, turn off. At a specified time in UTC, turn off a device.

ID::BP-PR-BTTS-010.002.003 On sunset, turn on. At sunset in UTC, turn on a device.

ID::BP-PR-BTTS-010.002.004 On sunset, turn off. At sunset in UTC, turn off a device.

ID::BP-PR-BTTS-010.002.005 On sunrise, turn on. At sunrise in UTC, turn on a device.

ID::BP-PR-BTTS-010.002.006 On sunrise, turn off. At sunrise in UTC,

turn off a device.

3.3.2 Gateway

ID::BP-PR-GW-010 The BP gateway shall support a web socket interface to clients such as http-server and timed event generator to trigger ZCL commands to bellows.

ID::BP-PR-GW-020 When started, the BP gateway shall determine the state of all devices (on, off, unknown) and record them for the purpose of passing the state information to clients.

ID::BP-PR-GW-030 When a client connects to the socket, the first message received shall be displayed to a log for the purpose of identifying what client connected.

ID::BP-PR-GW-040 When a client connects to the web socket, the gateway shall report to the client the status of each device supported.

ID::BP-PR-GW-050 When a client sends a command for bellows (any message following the initial connection message), the gateway shall verify that it is a proper command. All commands that identify an end-point in the ZigBee network using an ASCII IEEE Address with an 'on', 'off', or 'status' suffix will be valid.

ID::BP-PR-GW-060 When a client sends a command for bellows (any message following the initial connection message), the gateway shall log the command.

ID::BP-PR-GW-070 Bellows commands shall be passed to bellows for processing via an asyncio await command via the bellows API.

ID::BP-PR-GW-080 Bellows responses (device state updates - on/off/unknown), shall be passed back to all connected gateway clients.

ID::BP-PR-GW-090 If the gateway receives a message that is expected to be a bellows ZCL command but it is not valid (not the correct format or the IEEE Address is not defined), the gateway shall disconnect from the client.

3.3.3 HTTP-Server

ID::BP-PR-HS-010 The HTTP-Server shall support an HTTP server that supports a standard web browser.

ID::BP-PR-HS-020 The HTTP-Server shall be able to function (turn on/off devices) without JavaScript.

ID::BP-PR-HS-030 The HTTP-Server shall display buttons for each ZigBee controllable device (end-point) in the ZigBee network.

ID::BP-PR-HS-040 The HTTP-Server shall send bellows ZCL commands to the gateway when buttons are pressed.

ID::BP-PR-HS-050 The HTTP-Server shall make a web socket connection to the gateway.

ID::BP-PR-HS-060 The HTTP-Server shall, upon successful connection, send an identification to the gateway.

ID::BP-PR-HS-070 The HTTP-Server shall close the web socket connection when the HTTP connection to the browser is closed.

ID::BP-PR-HS-080 The HTTP-Server shall reflect device status from the gateway to the page being displayed in the browser.

3.3.4 Timed Event Generator

ID::BP-PR-TS-010 The Timed Event Generator shall use system time in UTC to turn on or off any configured devices.

ID::BP-PR-TS-020 The Timed Event Generator shall turn on individual devices at designated UTC times.

ID::BP-PR-TS-030 The Timed Event Generator shall turn off individual devices at designated UTC times.

ID::BP-PR-TS-040 Time designation shall be identified as:

ID::BP-PR-TS-040.001 A specific time of day.

ID::BP-PR-TS-040.002 The sunrise event plus an offset (+/-).

ID::BP-PR-TS-040.003 The sunset event plus an offset (+/-).

ID::BP-PR-TS-050 The Timed Event Generator shall make transient connections to the gateway via a web socket consisting of the following sequence:

ID::BP-PR-TS-050.001 Connect to the gateway.

ID::BP-PR-TS-050.002 Send an identification message.

ID::BP-PR-TS-050.003 Receive a status update.

ID::BP-PR-TS-050.004 Send a command message.

ID::BP-PR-TS-050.005 Receive a status update.

ID::BP-PR-TS-050.006 Close the connection.

3.4 Provisioning Requirements

ID::BP-PROV-010 BP ZigBee configuration shall use only information in the ZigBee database file.

ID::BP-PROV-021 BP sunrise and sunset configuration shall use only information in the sunset and sunrise configuration files.

3.5 Performance Requirements

ID::BP-PERF-GW-010 Bellows commands received from a gateway client shall be processed serially, in the order received, with no perceptible delay (less than a second) above the bellows delay.

ID::BP-PERF-HS-010 From button press, to display update, the time shall be no more than one second beyond the bellows processing time.

ID::BP-PERF-TS-010 From ZCL request, to status received, the time shall be no more than one second beyond the bellows processing time.